Metadata Concept and Semantic Quality Annotation Guidelines

Objective

The objective of this annotation task is to identify the concepts explicitly expressed in dataset metadata and provide a rating of the semantic relevance and semantic completeness of assigned keywords in relation to the descriptive metadata. The resulting annotations will serve as a human reference for evaluating the validity of automatically extracted concepts and the extent to which keyword annotations provide a conceptually complete and semantically relevant representation of metadata.

Dataset titles and descriptions must be annotated separately.

Definitions

Concept

A semantically coherent unit of meaning that is explicitly expressed in a metadata field and contributes to describing what the dataset is about or what it describes. Concepts may include topics, entities, activities, locations, events, populations, organizations, time periods, measurements, administrative units, scientific phenomena, technologies, and other domain-specific subjects.

Semantic Relevance

Semantic relevance is the extent to which a set of keywords is semantically related to the information expressed in descriptive metadata. A keyword set semantically aligns with a metadata field when the *meaning expressed* by the keywords corresponds to, and accurately reflects, information explicitly expressed in that metadata field.

Semantic Completeness

Semantic completeness is the extent to which the provided keywords semantically represent the important concepts explicitly expressed in the metadata. A keyword set has high semantic completeness when most or all-important concepts expressed in the metadata are adequately represented by one or more keywords. A concept is considered represented when one or more provided keywords convey the same or equivalent meaning as the concept expressed in the metadata. Exact lexical correspondence is not required; semantically equivalent or closely corresponding expressions may be considered representations.

Core Principles

1. Use Explicit Information Only

Concepts may be standardized or rephrased into meaningful, semantically equivalent terms, provided that their meaning is **explicitly expressed in the metadata**; annotators should not introduce new concepts through inference.

Metadata: *Replication Data for "Interval eccentric cycling: a promising training modality for sedentary people"*

- Concept: Sedentarity
  → represents the explicitly expressed concept *"sedentary people"* using a standard conceptual form.

Metadata: *Road accidents 2020*

- Concepts: Road accidents; 2020
- Traffic accidents → represents *Road accidents* through a semantically equivalent expression.
- Public safety → not explicitly expressed in the metadata.
- Traffic management → not explicitly expressed in the metadata.

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2. Prefer Semantic Concepts

Prefer meaningful phrases over isolated words whenever appropriate.

Good: Air pollution, public transport, Traffic accidents, Cystic fibrosis, Air quality monitoring

Bad: Air, Pollution, Transport, Traffic, Quality

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3. Preserve Compound Concepts

If multiple words form a single semantic concept, keep them together.

Good: Country of citizenship, Air quality monitoring, Road traffic accidents, European Train Control System, Public transport network

Bad: Country, Citizenship, Air, Quality, Monitoring

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4. Include Locations and Time Periods When Relevant

Locations and time periods may be annotated when they contribute to understanding the metadata.

Examples: Belgium, Brussels, Wallonia, North Sea, 2020, 2011 Census, 1990–2020

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5. Avoid Generic Concepts

Do not annotate generic metadata terms unless they represent a meaningful domain concept.

Avoid: Data, dataset, information, records, statistics

Avoid concepts that are trivial, overly generic, or unrelated to understanding the metadata.

Special Cases

Dataset Titles

When a dataset title contains multiple concepts, annotate the underlying concepts rather than the full dataset name.

Example

Metadata: Belgian Cystic Fibrosis Registry

Good: Belgium, Cystic fibrosis, Registry

Bad: Belgian Cystic Fibrosis Registry

Locations and Dates

Locations and dates should be annotated when they contribute to understanding the metadata.

Example

Metadata: Road accidents 2020

Good: Road accidents, 2020

Bad: Road accidents

Annotation Procedure

For each dataset:

Step 1: Annotate Title Concepts

- Read the dataset title.
- Identify between 1 and 5 concepts that best represent the meaning of the title.
- Record the concepts in order of importance.

Step 2: Annotate Description Concepts

- Read the dataset description.
- Identify between 3 and 8 concepts that best represent the meaning of the description.
- Record the concepts in order of importance.

Step 3: Assess Title Conceptual Completeness

- Review the keyword set provided for the dataset.

- Assess the extent to which the keywords capture the important concepts expressed in the title.
- Assign a rating using the scale provided in the Rating Guidelines.

Step 4: Assess Description Conceptual Completeness

- Assess the extent to which the keywords capture the important concepts expressed in the description.
- Assign a rating using the scale provided in the Rating Guidelines.

Step 5: Assess Title Semantic Relevance

- Assess the extent to which the provided keywords are semantically relevant to, and accurately reflect, the information expressed in the title.
- Assign a rating using the scale provided in the Rating Guidelines.

Step 6: Assess Description Semantic Relevance

- Assess the extent to which the provided keywords are semantically relevant to, and accurately reflect, the information expressed in the description.
- Assign a rating using the scale provided in the Rating Guidelines.

Rating Guidelines

Semantic Relevance

Level	Definition
Low Relevance	Little or no meaningful representation of information expressed in metadata.
Medium Relevance	The keyword set is meaningfully related to the metadata but represents it only broadly, partially, or indirectly.
High Relevance	The keywords directly and accurately represent information explicitly expressed in the metadata.

Semantic Completeness

Level	Definition
Low Completeness	Important concepts are largely absent from the keywords set.
Medium Completeness	The keywords capture a substantial but incomplete portion of the semantic content of the concepts. Some important concepts are represented, while others are missing.
High Completeness	Nearly all-important concepts are represented.

For all completeness and relevance assessments, evaluate the keywords only against the information contained in the title or description being assessed. Do not use external knowledge about the dataset.

Consistency Guidelines

Use metadata wording where possible, use consistent terminology, prefer meaningful concept phrases, focus on concepts central to understanding the metadata, avoid duplicates, and annotate only explicitly stated concepts.

For consistency across annotators, identify between 1 and 5 concepts for dataset titles and between 3 and 8 concepts for dataset descriptions. If no meaningful concept is explicitly expressed in the title or description, enter **"None"** in the corresponding concept field rather than leaving it blank.

Annotation File Instructions

Annotators will receive a spreadsheet (CSV or Excel file) containing approximately 300 datasets sampled from the European Open Data Portal, including the following fields:

• Dataset ID, Dataset Title, Dataset Description, Dataset Keywords

Six annotation columns must be completed:

- **title_concepts:** Enter the concepts identified in the dataset title.
- **description_concepts:** Enter the concepts identified in the dataset description.
- **title_completeness_rating**: Enter a rating indicating how well the provided keywords capture the important concepts expressed in the dataset title.
- **description_completeness_rating:** Enter a rating indicating how well the provided keywords capture the important concepts expressed in the dataset description.
- **title_relevance_rating:** Enter a rating indicating how semantically relevant the provided keywords are to the information expressed in the dataset title.
- **description_relevance_rating:** Enter a rating indicating how semantically relevant the provided keywords are to the information expressed in the dataset description.

Formatting Rules

Enter concepts as a semicolon-separated list.

Example: Air pollution; Monitoring stations; Belgium
